

The Training-Bound Emergence of Bayesian Hypothesis Elimination in Transformer Attention

An Information-Theoretic and Geometric Study in Bijection Wind Tunnels

Advanced Topics in Physics of Data

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Abstract

We reproduce and extend the bijection elimination Bayesian wind tunnel from Agarwal et al. (2026), a synthetic benchmark where a Transformer must track an analytically known posterior over a hidden bijection. Motivated by a possible inconsistency between the reference paper’s stated methodology and its reported results, we introduce an experimental design varying query horizons and training repetition regimes. We find that, when evaluated with out-of-distribution repeated queries (following Agarwal et al.), only models exposed to repetitions distributed across the entire sequence during training (Models A and C) successfully track the exact Bayesian posterior predictive distribution and develop clean Bayesian geometry (orthogonal keys and an unfurled value manifold). In contrast, models never exposed to early repetitions during training (Models B and D) fail, either adopting a rigid positional shortcut or failing to develop associative retrieval altogether. However, when evaluated without repeated queries, all models perform well. Our results provide strong evidence that Bayesian-like reasoning in Transformers is not guaranteed to generalize beyond the training distribution, despite being optimal on the training distribution under the population cross-entropy objective.

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1 Introduction: Transformers in Bayesian Wind Tunnels

Transformers constitute the foundational architecture behind modern Large Language Models (LLMs). A prominent theoretical perspective interprets their execution as implicit Bayesian inference. However, determining whether Transformers perform probabilistic reasoning is constrained by the fact that natural language data offer no analytic ground truth. Even when a model *behaves* as if it were updating beliefs Bayesianly, it is difficult to demonstrate that this pass is actually Bayesian, as the true posterior for LLMs is mathematically intractable.

Bayesian wind tunnels are controlled synthetic benchmarks designed to resolve this barrier. In these environments, the data generating process is fully specified and ensures that the exact Bayesian posterior is known analytically at every step for any variable. Simultaneously, the hypothesis space is chosen to be vast, making direct memorization impractical and forcing the network to implement probabilistic reasoning.

This project focuses on the **bijection elimination** task, originally formulated in Agarwal et al. 2026 [1] to investigate the *belief accumulation* primitive of Bayesian inference. An optimal model performing belief accumulation must sequentially integrate incoming evidence into a running posterior $P(\theta \mid \{(x_i, y_i)\}_{i=1}^k)$. This process corresponds directly to hypothesis elimination, where inconsistent mappings given the observations are pruned via Bayes' theorem.

The paper explains the Bayesian behavior of Transformers in wind tunnels through a geometric argument. Across the attention layers, the authors propose a three-stage mechanism:

- **Foundational Binding (First Attention Layer):** Construct an orthogonal hypothesis frame parameterized by key vectors.
- **Progressive Elimination (Middle Attention Layers):** Sequentially suppress inconsistent hypotheses through sharpening QK alignment.
- **Precision Refinement (Late Attention Layers):** Encode posterior entropy on a smooth value manifold while keeping the orthogonal frame fixed.

This structure mirrors the analytic structure of Bayesian hypothesis elimination: define a hypothesis space, update beliefs with evidence, and finally refine confidence as uncertainty decreases.

The core objectives of this report are:

- **Replication of the Wind Tunnel:** Implement and train standard autoregressive Transformers (minGPT) on the bijection task to verify whether attention heads learn to track the exact Bayesian predictive distribution and its entropy.
- **Resolution of Literature Inconsistency:** Probe an apparent contradiction in the reference paper [1] between its formal problem definition (stating inputs never repeat) and its reported behavior (displaying an entropy collapse upon repeated inputs), by contrasting Transformers trained with different distribution.
- **Mechanistic and Geometric Characterization:** Track the evaluation metrics and the geometric diagnostics proposed by the reference paper (key orthogonality, QK sharpening, value-manifold unfurling) across the entire training trajectory for all the models.

2 Theoretical Background

2.1 Cross-Entropy and Gibbs' Inequality

Cross-entropy measures the average coding cost (in bits or nats) required to describe outcomes sampled from a true data distribution p when using an optimized codebook parameterized by a

model distribution q :

$$H(p, q) = \mathbb{E}_{y \sim p}[-\log q(y)] = -\sum_{y \in \mathcal{Y}} p(y) \log q(y). \quad (1)$$

The term $-\log q(y)$ quantifies the individual penalty incurred when the realization $y \sim p(y)$ occurs under the predictive distribution q .

Decomposing the logarithm inside the cross-entropy definition yields:

$$H(p, q) = -\sum_{y \in \mathcal{Y}} p(y) \log q(y) = -\sum_{y \in \mathcal{Y}} p(y) \log \left[p(y) \frac{q(y)}{p(y)} \right] \quad (2)$$

$$= -\sum_{y \in \mathcal{Y}} p(y) \log p(y) - \sum_{y \in \mathcal{Y}} p(y) \log \frac{q(y)}{p(y)} \quad (3)$$

$$= H(p) + D_{\text{KL}}(p \parallel q), \quad (4)$$

where $D_{\text{KL}}(p \parallel q) \geq 0$ is the Kullback–Leibler (KL) divergence, strictly non-negative by Jensen’s inequality. Consequently,

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q) \geq H(p), \quad (5)$$

with equality holding if and only if $q(y) = p(y)$ for all $y \in \mathcal{Y}$.

Here, $H(p) = -\sum_y p(y) \log p(y)$ denotes the Shannon entropy of p , representing the irreducible uncertainty inherent to the source.

2.2 Population Optimum of Cross-Entropy

Setup. Let latent parameters be drawn from a prior $\theta \sim \pi(\theta)$. Given a context sequence $c = \{(x_i, y_i)\}_{i=1}^{k-1}$ and a current query input x , labels y are generated according to the conditional likelihood $p(y \mid x; \theta)$. A parameterized predictive model $q(y \mid x, c)$ is optimized by minimizing the population cross-entropy loss:

$$\mathcal{L}(q) = \mathbb{E}_{\theta \sim \pi} \mathbb{E}_{c, (x, y) \sim p(\cdot \mid \theta)}[-\log q(y \mid x, c)]. \quad (6)$$

Theorem 1. *The global minimizer $q^*(y \mid x, c)$ of the population cross-entropy $\mathcal{L}(q)$ is uniquely given by the Bayesian posterior predictive distribution:*

$$q^*(y \mid x, c) = \int p(y \mid x, \theta) p(\theta \mid c) d\theta, \quad (7)$$

where $p(\theta \mid c)$ is the exact Bayesian posterior:

$$p(\theta \mid c) \propto \pi(\theta) \prod_{(x_i, y_i) \in c} p(y_i \mid x_i; \theta). \quad (8)$$

Proof. We assume the following independencies:

1. $(y \perp c) \mid (x, \theta)$, meaning x and θ render the label conditionally independent of the rest of the context;
2. $(x \perp \theta) \mid c$, meaning the context c fully accounts for any information about θ carried by query x .

Under these assumptions, marginalizing θ expresses the loss as an expectation over the joint distribution $p(x, c)$ and the true posterior predictive $p(y \mid x, c)$:

$$\mathcal{L}(q) = \mathbb{E}_{(x,c)} \left[\mathbb{E}_{y \sim p(y|x,c)} \left[-\log q(y \mid x, c) \right] \right]. \quad (9)$$

Because $q(\cdot \mid x, c)$ may be chosen independently for each pair (x, c) , minimizing $\mathcal{L}(q)$ is equivalent to pointwise minimization of the inner cross-entropy for every (x, c) :

$$q^*(\cdot \mid x, c) = \underset{q}{\operatorname{argmin}} \mathbb{E}_{y \sim p(y|x,c)} \left[-\log q(y \mid x, c) \right]. \quad (10)$$

Applying Gibbs' inequality to the conditional distributions yields:

$$\mathbb{E}_{y \sim p(y|x,c)} \left[-\log q(y \mid x, c) \right] = H(p(\cdot \mid x, c)) + D_{\text{KL}}(p(\cdot \mid x, c) \parallel q(\cdot \mid x, c)) \geq H(p(\cdot \mid x, c)). \quad (11)$$

The lower bound is achieved if and only if $D_{\text{KL}}(p \parallel q) = 0$, requiring:

$$q^*(y \mid x, c) = p(y \mid x, c). \quad (12)$$

Conditioning on the latent parameter θ and integrating over the parameter posterior, we obtain:

$$q^*(y \mid x, c) = \int p(y \mid x, c, \theta) p(\theta \mid x, c) d\theta = \int p(y \mid x, \theta) p(\theta \mid c) d\theta, \quad (13)$$

where we have utilized the independencies stated above. $\square$

Remark 1. *This result establishes that a model with unlimited capacity trained under cross-entropy converges to the Bayesian predictive posterior. Notably, this convergence is strictly valid with respect to the data distribution sampled during training.*

2.3 The Bijection Elimination Task

The *bijection learning* (or bijection elimination) task instantiates wind tunnel formalism in a discrete combinatorial domain. Let $\mathcal{V} = \{0, 1, \dots, V-1\}$ be a finite discrete vocabulary of size V . The latent parameter θ corresponds to a uniformly sampled random bijection $\pi : \mathcal{V} \rightarrow \mathcal{V}$. Memorization is infeasible since the hypothesis space size is $V!$.

A sequence of length K consists of successive input-output pairs. At query step $k \in \{1, \dots, K\}$, the model is presented with a context of $k-1$ previously revealed pairs:

$$c_{k-1} = \{(x_1, y_1), (x_2, y_2), \dots, (x_{k-1}, y_{k-1})\}, \quad \text{with } y_i = \pi(x_i). \quad (14)$$

Let $\mathcal{X}_{k-1} = \{x_1, \dots, x_{k-1}\}$ and $\mathcal{Y}_{k-1} = \{y_1, \dots, y_{k-1}\}$ denote the sets of inputs and outputs observed prior to step k , and let n_{used} be the number of distinct mappings revealed so far.

Given the k -th query input x_k , the Bayesian posterior predictive distribution $p(y \mid x_k, c_{k-1})$ partitions cleanly into two distinct operational regimes:

1. **Repeated Query (Memorization):** If the query input has already appeared in the context (i.e., $x_k = x_i \in \mathcal{X}_{k-1}$ for some $i < k$), then the output must match the earlier observation. The posterior collapses to a deterministic Dirac delta mass:

$$p(y \mid x_k, c_{k-1}) = \delta_{y, y_i}. \quad (15)$$

The exact posterior entropy is identically zero:

$$H_{\text{Bayes}}(k) = 0 \text{ bits}. \quad (16)$$

2. **Novel Query (Hypothesis Elimination):** If the query input has not been observed previously ($x_k \notin \mathcal{X}_{k-1}$), because the prior over unobserved bijections is uniform, then the posterior remains uniform over the remaining $V - n_{\text{used}}$ available labels:

$$p(y \mid x_k, c_{k-1}) = \begin{cases} \frac{1}{V - n_{\text{used}}} & \text{if } y \in \mathcal{V} \setminus \mathcal{Y}_{k-1}, \\ 0 & \text{if } y \in \mathcal{Y}_{k-1}. \end{cases} \quad (17)$$

The exact posterior entropy is given by:

$$H_{\text{Bayes}}(k) = \log_2(V - n_{\text{used}}) \text{ bits.} \quad (18)$$

In the special case where all previous $k-1$ queries were distinct ($n_{\text{used}} = k-1$), the entropy simplifies to:

$$H_{\text{Bayes}}(k) = \log_2(V - k + 1) \text{ bits.} \quad (19)$$

To minimize cross-entropy, a Transformer must dynamically suppress the probabilities corresponding to forbidden tokens $\mathcal{Y}_{k-1}$ while maintaining uniform probabilities over the set $\mathcal{V} \setminus \mathcal{Y}_{k-1}$.

We note that the reference paper’s own implementation of this task fixes $V = 20$ and, by explicitly constraining every input to appear *at most once* per sequence, uses a query horizon of $K = V - 1 = 19$: this guarantees that $n_{\text{used}} < V$ at every step, so that a repeated query can never structurally occur. In this project, we use this setup and also a different query horizon of $K = 1.5V$, as described in Section 3.2.

2.4 Bayesian Geometric Mechanism in Transformers

Evidence from Agarwal et al. [1] suggests that Transformers perform Bayesian inference by constructing a geometric hypothesis frame, executing sequential elimination within it, and finally refining posterior precision across layers.

The computation starts in the first Attention Layer, in which a hypothesis space is built to serve as the foundation for subsequent inference steps. Here, key vectors form an approximately orthogonal basis, providing a coordinate system where probability mass can be geometrically represented and manipulated. This orthogonality is measured via the mean absolute off-diagonal cosine similarity between key vectors k_i and k_j :

$$\mathcal{M} = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{j \neq i} \left| \frac{k_i^\top k_j}{\|k_i\|_2 \|k_j\|_2} \right| \quad (20)$$

where N is the number of key vectors. This hypothesis space emerges at early checkpoints in training and remains fairly stable throughout the rest of the training process.

Within this frame, the middle Attention Layers align query vectors q_i more strongly with the subset of keys consistent with the observed evidence. While early layers attend broadly, deeper layers concentrate attention almost exclusively on the feasible hypotheses. This geometric QK sharpening therefore resembles analytic Bayesian conditioning. Mathematically, for each attention head h we compute the dot product between the normalized query $\hat{q}_t^{(h)}$ and key $\hat{k}_j^{(h)}$ vectors, yielding a cosine similarity between query position t and key position j :

$$c_{t,j}^{(h)} = \hat{q}_t^{(h)\top} \hat{k}_j^{(h)} = \sum_{m=1}^{d_{\text{head}}} \hat{q}_{t,m}^{(h)} \hat{k}_{j,m}^{(h)}$$

We average this score element-wise across the H heads to obtain a single scalar alignment score $c_{t,j}$ for each pair of positions:

$$c_{t,j} = \frac{1}{H} \sum_{h=1}^H c_{t,j}^{(h)}$$

Then, to isolate the relevant hypothesis space, we apply a hypothesis mask to isolate only the input queries and perform a row-wise softmax restricted over the valid key positions $c_{t,\cdot}$, which yields a distribution $r_{t,j}$ over hypotheses. The entropy of this distribution (in nats) quantifies how concentrated the alignment is: a lower entropy indicates that the query attends sharply to a small subset of keys, while a higher entropy indicates diffuse attention across many possibilities.

Finally, the deepest layers refine the precision of the posterior representation. To geometrically evaluate this, the block attention output vectors $z_t \in \mathbb{R}^{d_{model}}$ from the final layer are extracted at the prediction positions across 200 different bijections evaluated in the wind tunnel framework. Each position is associated with its analytic Bayesian entropy H_{Bayes} defined in Equation 18. By applying Principal Component Analysis (PCA) to project these high-dimensional representations into a 3D space, two distinct behaviors are observed. At early checkpoints, value representations of low-entropy states are nearly collapsed together, failing to geometrically distinguish varying levels of certainty. Conversely, at final checkpoints, the representation space becomes more regular: the embedded states distribute along a smooth manifold, where a point’s position is directly parameterized by its posterior entropy H_{Bayes} .

3 Experiments

3.1 Implementation and Evaluation Metrics

We implemented the bijection wind tunnel using a 6-layer, 6-head, decoder-only Transformer (minGPT [2], 2.69M parameters, embedding dimension $d = 192$) closely matching the dimensions of the Transformer trained in the reference paper [1]. We trained our Transformers on NVIDIA T4 GPUs using Kaggle infrastructure. Sequences were tokenized as flat lists

$$[x_1, y_1, \text{SEP}, x_2, y_2, \text{SEP}, \dots, x_K, y_K, \text{SEP}]$$

of length $3K$. Following the standard next-token training setup, the model receives this sequence with the final token dropped (length $3K - 1$) as input, and the same sequence shifted by one position as target; loss masking was then applied to evaluate cross-entropy exclusively at output prediction positions $p = 3(k - 1)$ for $k \in \{1, \dots, K\}$.

To assess whether the model tracks the true distribution, model predictions $q(y)$ were evaluated across four metrics:

- **Mean Absolute Entropy Error (MAE):** Evaluated on novel queries to measure calibration against the analytic uniform entropy:

$$\text{MAE} = \frac{1}{N_{\text{new}}} \sum_{k \in \mathcal{K}_{\text{new}}} |H_{\text{model}}(k) - H_{\text{Bayes}}(k)| \quad (21)$$

where $\mathcal{K}_{\text{new}}$ is the set of positions corresponding to novel queries, with $N_{\text{new}} = |\mathcal{K}_{\text{new}}|$

- **Mean Kullback–Leibler Divergence (D_{KL}):** Measuring full distributional divergence from the exact Bayesian uniform target over unclaimed tokens $\mathcal{U}_k = \mathcal{V} \setminus \mathcal{Y}_{k-1}$ (where $M = |\mathcal{U}_k|$). Because p_{Bayes} is perfectly uniform over the M available options and zero elsewhere, the general sum reduces exclusively to the $\mathcal{U}_k$ domain. By expanding the logarithm, we

find:

$$\begin{aligned}
D_{\text{KL}}(p_{\text{Bayes}} \parallel q_{\text{model}}) &= \sum_{y \in \mathcal{V}} p_{\text{Bayes}}(y) \log_2 \left(\frac{p_{\text{Bayes}}(y)}{q_{\text{model}}(y)} \right) \\
&= \sum_{y \in \mathcal{U}_k} \frac{1}{M} \log_2 \left(\frac{1/M}{q_{\text{model}}(y)} \right) \\
&= -\log_2 M - \frac{1}{M} \sum_{y \in \mathcal{U}_k} \log_2 q_{\text{model}}(y). \tag{22}
\end{aligned}$$

- **Forbidden Probability Mass:** The cumulative probability assigned to already-claimed outputs during novel queries, which should strictly vanish under optimal hypothesis elimination:

$$\text{MF}(k) = \sum_{y \in \mathcal{Y}_{k-1}} q_{\text{model}}(y \mid x_k, c_{k-1}). \tag{23}$$

- **Repetition Accuracy** $P(y_{\text{true}} \mid \text{repeat})$: The probability assigned to the correct mapping when an input repeats, which must approach 1.0 under exact associative retrieval.

3.2 Experimental Design

During task reproduction, an apparent discrepancy was identified in the paper [1], which persists unchanged across all arXiv revisions up to v5 (16 May 2026). In Sections 2.3 and 3.1, the setup explicitly specifies that "inputs never repeat" within a sequence, a claim implemented by fixing the query horizon to $K = V - 1 = 19$ for $V = 20$ and the probability of query repetition in a sequence to zero ($p_{\text{repeat}, \text{train}} = 0$). Since at most $V - 1$ of the V available inputs are ever queried, $n_{\text{used}} < V$ holds at every step. In contrast, in Figure 4 of the paper they also analyze an entropy trajectory on different test bijections in which the entropy repeatedly collapses to 0 "when previously seen inputs reappear". However, as stated before, repeated queries are impossible under the paper's own $K = V - 1$ construction, so such a sawtooth trajectory cannot arise from correctly tracking the analytic posterior on the training distribution as declared. One might argue instead that the model has learned a behavior that generalizes beyond the training support to handle repeated queries correctly. However, the population-optimum theorem of Section 2.2 fixes the optimal predictor only with respect to the distribution sampled during training, and offers no guarantee about model behavior on inputs outside that support.

Several explanations for this discrepancy are possible and cannot be distinguished from the text alone. We instead treat the discrepancy as motivation for an independent empirical test, where we train four different models under two query horizons and two repetition regimes:

- **Model A:** $K = 1.5V = 30$, trained with a repetition probability $p_{\text{repeat}, \text{train}} = 0.3$.
- **Model B:** $K = 1.5V = 30$, trained strictly with $p_{\text{repeat}, \text{train}} = 0.0$, thus no early repeated queries are allowed.
- **Model C:** $K = V - 1 = 19$, reproducing the reference paper's own query horizon, trained with $p_{\text{repeat}, \text{train}} = 0.3$.
- **Model D:** $K = V - 1 = 19$, reproducing the reference paper's own query horizon, trained with $p_{\text{repeat}, \text{train}} = 0.0$, exactly matching the setup declared in the reference paper.

The choice of query horizon has a direct consequence on whether repeated queries can occur at all. For Models A and B, $K = 1.5V$ exceeds the vocabulary size. In particular for Model B, once all $V = 20$ inputs have been queried, every remaining step ($k = 21, \dots, 30$) is a forced repeat, since no unused input remains to sample. Model B is therefore never exposed to early

repeats during training, but it encounters late repeats in every single sequence. Instead, Model A both encounters early and late repetitions. By contrast, for Models C and D, $K = V - 1 = 19$ guarantees $n_{\text{used}} < V$ at every step. Model D is the only model completely isolated from repeated queries in any form, perfectly reproducing the training distribution declared in the reference paper, whereas Model C still encounters repeats in the sequence.

This design provides a controlled 2×2 comparison of query horizon and repetition exposure. Models A and B test the effect of deliberate early repetition exposure while holding fixed a training distribution that guarantees late repeats. Comparing them reveals whether a model, which is exposed to abundant and positionally fixed late repeats, can construct a position-invariant retrieval mechanism for early repetitions. Models C and D instead directly replicate the reference paper’s query horizon, removing forced late repeats entirely. Specifically, Model C establishes whether the geometric mechanism observed in Model A is an artifact of the extended horizon $K = 1.5V$, while Model D provides the most direct test of the reference paper’s stated methodology. As formalized in Section 4, this design is central to interpreting the results across models, and finding if generalizing behavior can be constrained by Theorem 1.

All models are trained for a total of 140,000 iterations. To effectively capture the training dynamics in Models A and C, which show sudden changes in metrics as phase transitions, we use an adaptive checkpointing schedule: weights are saved every 500 steps during the first 20,000 iterations, and every 10,000 steps thereafter. This identical schedule is applied to Models B and D for consistency. Although the training loss plateaus relatively early, we deliberately extend the training regime to approach the timescale of the original study ($\sim 150,000$ iterations) and to observe possible late phase transitions in the attention mechanism that remain untracked by standard loss curves.

All models were optimized using AdamW with a batch size of 64 sequences. We employed a constant learning rate of 10^{-3} . Weight decay was set to 0.1, and gradient clipping was applied at a maximum norm of 1.0.

All four models were subsequently subjected to an evaluation benchmark with

$$p_{\text{repeat,eval}} \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$$

to properly analyze their in and out-of-distribution performance.

Parameter	Model A	Model B	Model C	Model D
$p_{\text{repeat,train}}$	0.3	0.0	0.3	0.0
Query Horizon (K)	$1.5V = 30$	$1.5V = 30$	$V - 1 = 19$	$V - 1 = 19$

Table 1: Summary of the 2×2 experimental design comparing query horizon and repetition exposure.

4 Results

We report calibration metrics and geometric diagnostics for all four models across the full 140,000 iteration training, evaluated on 2000 different bijections. The final calibration metrics, evaluated across the full $p_{\text{repeat,eval}}$ sweep, are summarized in Tables 2–5, while the geometric diagnostics at key training checkpoints are in Tables 6–9.

When $p_{\text{repeat,eval}} = 0$, all models at iteration 140,000 perform Bayesian-like inference on this in-distribution regime, consistent with Theorem 1 (note that for Models C and D, $P(y_{\text{true}} \mid \text{repeat})$ is undefined since no repetitions of any kind ever occur in this setup). Therefore, to capture

out-of-distribution behaviors and reproducing the evaluation setup behind Agarwal’s entropy sawtooth pattern, the following discussion focuses on evaluation sets whose probabilities of query repetitions are strictly greater than zero. Without loss of generality, we examine $p_{\text{repeat,eval}} = 0.3$. Model checkpoints and full metrics are available at the Github repository [3].

Table 2: Full calibration metrics for Model A ($K = 1.5V$, $p_{\text{repeat,train}} = 0.3$) at iteration 140,000.

$p_{\text{repeat,eval}}$	MAE(new) [bits]	D_{KL} [bits]	Forbidden mass	$P(y_{\text{true}} \text{repeat})$
0.0	0.014	0.021	0.003	0.9999
0.1	0.014	0.021	0.003	0.9999
0.2	0.014	0.021	0.003	0.9999
0.3	0.014	0.022	0.003	0.9999
0.4	0.014	0.021	0.003	0.9999
0.5	0.013	0.021	0.003	0.9999
0.6	0.012	0.019	0.003	0.9999

Table 3: Full calibration metrics for Model B ($K = 1.5V$, $p_{\text{repeat,train}} = 0.0$) at iteration 140,000.

$p_{\text{repeat,eval}}$	MAE(new) [bits]	D_{KL} [bits]	Forbidden mass	$P(y_{\text{true}} \text{repeat})$
0.0	0.010	0.019	0.002	0.9999
0.1	0.105	1.531	0.095	0.7200
0.2	0.257	2.969	0.190	0.4993
0.3	0.448	3.926	0.266	0.3498
0.4	0.626	4.312	0.304	0.2752
0.5	0.769	4.157	0.310	0.2592
0.6	0.878	3.796	0.302	0.2529

Table 4: Full calibration metrics for Model C ($K = V - 1$, $p_{\text{repeat,train}} = 0.3$) at iteration 140,000.

$p_{\text{repeat,eval}}$	MAE(new) [bits]	D_{KL} [bits]	Forbidden mass	$P(y_{\text{true}} \text{repeat})$
0.0	0.028	0.022	0.004	—
0.1	0.020	0.020	0.004	0.9997
0.2	0.015	0.019	0.003	0.9999
0.3	0.013	0.017	0.003	0.9998
0.4	0.011	0.016	0.002	0.9999
0.5	0.010	0.014	0.002	0.9998
0.6	0.008	0.012	0.002	0.9999

Table 5: Full calibration metrics for Model D ($K = V - 1$, $p_{\text{repeat,train}} = 0.0$) at iteration 140,000.

$p_{\text{repeat,eval}}$	MAE(new) [bits]	D_{KL} [bits]	Forbidden mass	$P(y_{\text{true}} \text{repeat})$
0.0	0.012	0.018	0.002	—
0.1	0.011	0.017	0.002	0.0004
0.2	0.010	0.017	0.002	0.0004
0.3	0.009	0.017	0.002	0.0004
0.4	0.009	0.018	0.002	0.0005
0.5	0.009	0.019	0.002	0.0005
0.6	0.010	0.022	0.002	0.0007

4.1 Calibration Dynamics over Training

Figures 1 and 2 track four calibration metrics against training iteration for the four models evaluated at $p_{\text{repeat},\text{eval}} = 0.3$: new-query entropy error ($\text{MAE}(\text{new})$), Kullback-Leibler divergence (D_{KL}), forbidden probability mass on claimed outputs, and repeat-query calibration $P(y_{\text{true}} \mid \text{repeat})$.

Model A and Model C demonstrate remarkably similar and successful learning trajectories despite their different query horizons. Both undergo a rapid phase transition in $P(y_{\text{true}} \mid \text{repeat})$ between roughly iteration 5,000 and 20,000, after which the metric saturates near 1.0 and remains stable through iteration 140,000. Forbidden probability mass mirrors this behavior in reverse, collapsing to near-zero. Moreover, both $\text{MAE}(\text{new})$ and D_{KL} drop to the order of 10^{-2} bits within the first few thousand iterations. This confirms that exposure to probabilistic repeats during training reliably instills the complete Bayesian algorithm, regardless of the sequence horizon.

In contrast, the models B and D diverge entirely from them and from each other, exposing the fragility of out-of-distribution generalization.

Model B exhibits a trajectory driven by forced late repeats. For the first $\sim 50,000$ iterations, $P(y_{\text{true}} \mid \text{repeat})$ remains flat at approximately chance level ($1/20 \approx 0.03\text{--}0.05$). A transition occurs at iteration 60,000, jumping to a plateau around 0.32–0.35, while forbidden mass stabilizes around 0.22–0.26. Crucially, tracking D_{KL} reveals the underlying degradation: while MAE drops from ≈ 0.7 to $\approx 0.4\text{--}0.5$ bits, the D_{KL} trajectory explodes to 2.0–4.0 bits immediately after the transition. This divergence demonstrates that Model B becomes overconfident in incorrect answers for novel queries, adopting a positional heuristic rather than a valid Bayesian algorithm.

Completely isolated from any repeats during training, Model D never develops the degenerate shortcut seen in Model B. Instead, it becomes a perfect optimizer for hypothesis elimination on novel queries. Its $\text{MAE}(\text{new})$ and D_{KL} drop to $\sim 10^{-2}$ bits even faster than Model C, and its forbidden mass successfully approaches zero. However, it fails on out-of-distribution repeated queries. Because Model D learns the rule to suppress the probability of already-seen tokens, it actively assigns near-zero probability to the correct associative answer when forced to evaluate a repeat. Consequently, its $P(y_{\text{true}} \mid \text{repeat})$ remains flat near 0.0004 throughout training.

This contrast highlights an important distinction between in-distribution predictive optimality and algorithmic generalization: Model D accurately implements hypothesis elimination on the training support while failing to extend this computation to the structurally different repeated-query regime.

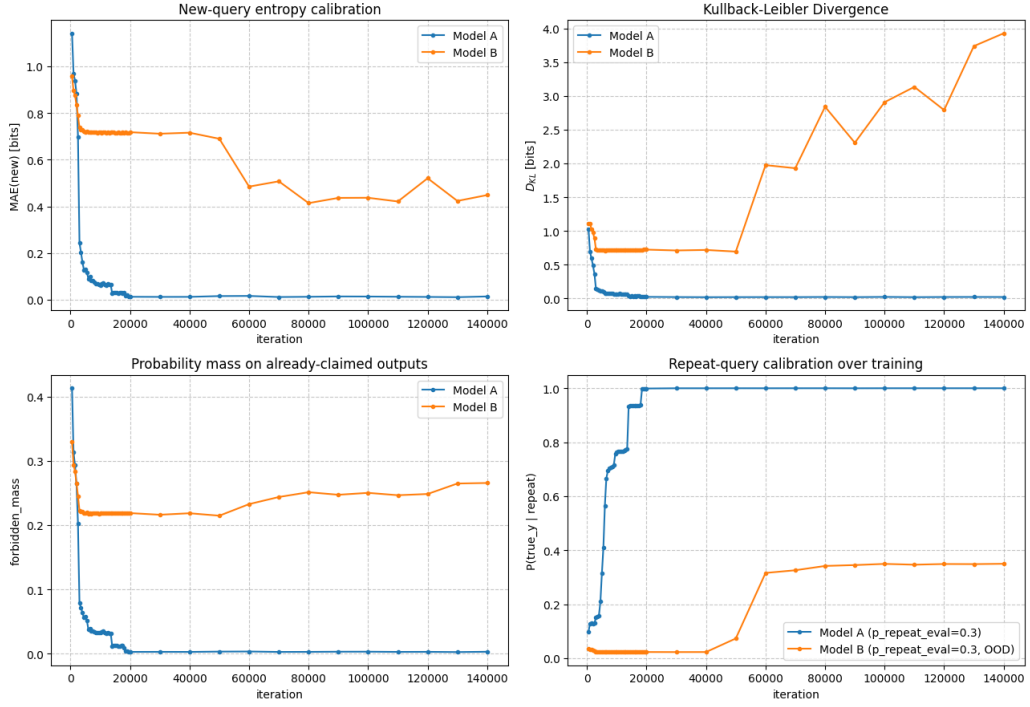


Figure 1: Training dynamics for Model A and Model B evaluated at $p_{repeat,eval} = 0.3$: new-query entropy error (MAE(new)), Kullback-Leibler divergence (D_{KL}), forbidden probability mass on claimed outputs, and repeat-query calibration $P(y_{true} | repeat)$.

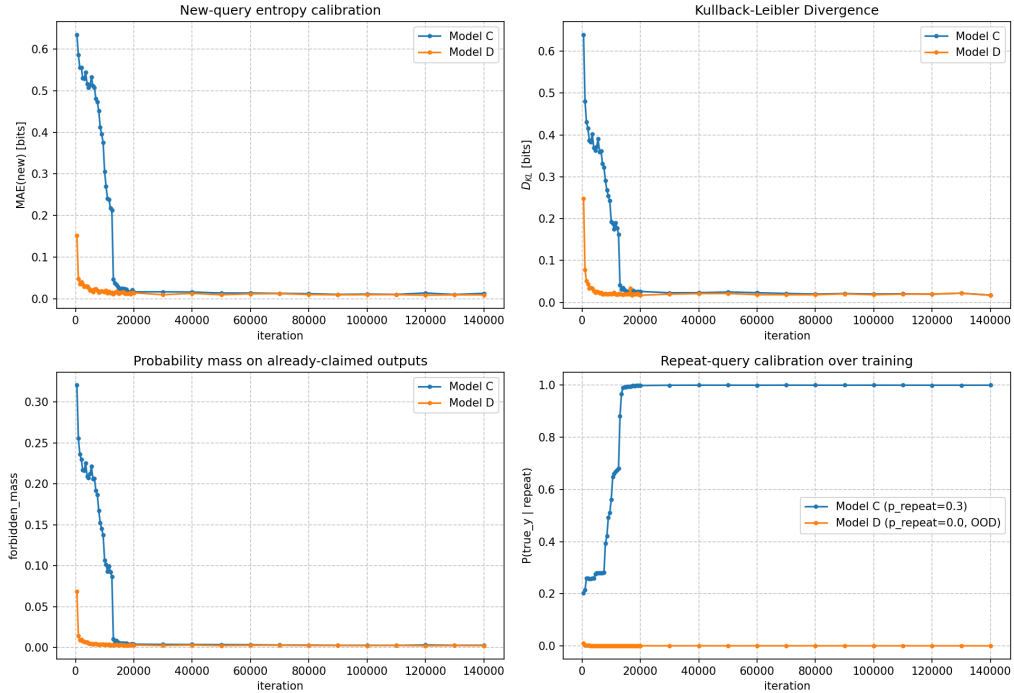


Figure 2: Training dynamics for Model C and Model D evaluated at $p_{repeat,eval} = 0.3$. Model D rapidly drives D_{KL} to zero but categorically fails on associative retrieval, as shown by its flat $P(y_{true} | repeat)$.

4.2 Per-Sequence Entropy Trajectories

We analyze the entropy trajectories of individual sequences, mirroring the diagnostic approach of Agarwal et al. [1] (see their paper, Figure 4).

For Models A and C at the final checkpoint, the predicted entropy perfectly traces the exact Bayesian posterior predictive distribution. Every time a query repeats, the model’s entropy collapses deterministically to zero. This confirms that training with repeats instills a position-invariant associative retrieval, regardless of the sequence horizon.

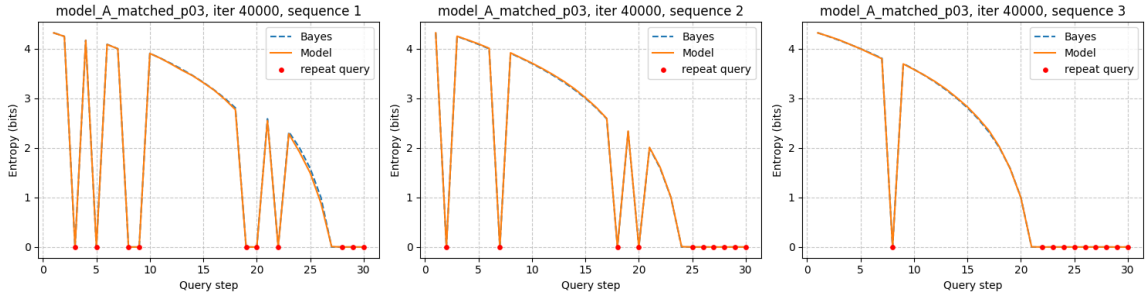


Figure 3: Per-sequence entropy trajectories for Model A at iteration 140,000 evaluated at $p_{\text{repeat,eval}} = 0.3$. The model matches the Bayesian posterior exactly.

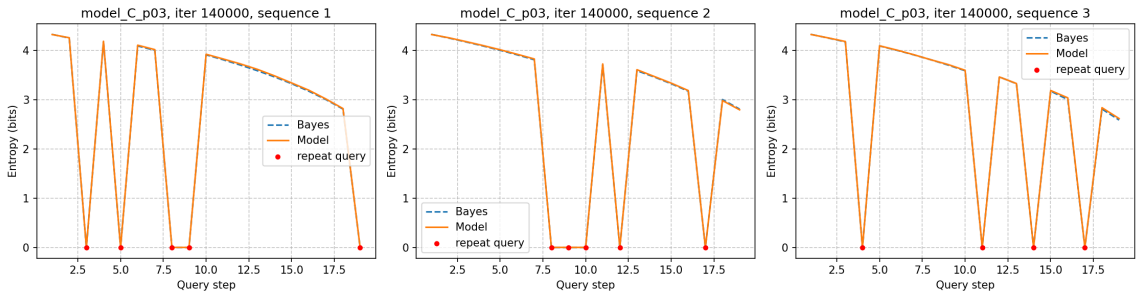


Figure 4: Per-sequence entropy trajectories for Model C at iteration 140,000. Similar to Model A, it perfectly tracks the Bayesian posterior.

Models B and D, however, reveal their failure in out-of-distribution scenarios, which becomes clear by comparing its state before its transition (iteration 40,000, Figure 5) with its final state (iteration 140,000, Figure 6).

Before the transition, Model B operates under a failing heuristic. For the first 20 steps, it ignores deliberate repeated queries, and after step 20 the entropy reaches the maximum possible value (~ 4.3 bits). At this stage, the model simply outputs a uniform distribution when faced with forced late repeats. After the transition, Model B learns a partial copying mechanism. It still ignores the deliberate early repeats, treating them as novel queries. However, starting from step 21, the entropy suddenly collapses toward zero, as the model has learned to copy the queries looking the past steps. Nevertheless, the copying is unstable, showing chaotic spikes in entropy rather than the clean, flat zero seen in models A and C.

This behavior perfectly illustrates the boundaries of Theorem 1, and the two checkpoints reveal how gradient descent exploits the training distribution’s structural constraints. By iteration 140,000, optimization dynamics force the model to minimize the loss on late repeats, leading it to adopt a rigid positional shortcut that collapses its entropy only from the 21st step onward. This positional trick explains Model B’s ≈ 0.32 – 0.35 plateau in $P(y_{\text{true}} \mid \text{repeat})$ (Section 4.1).

Because the heuristic activates deterministically only late in the sequence, the plateau reflects this artifact ($10/30 \approx 33.3\%$ of the sequence) rather than true position-invariant retrieval.

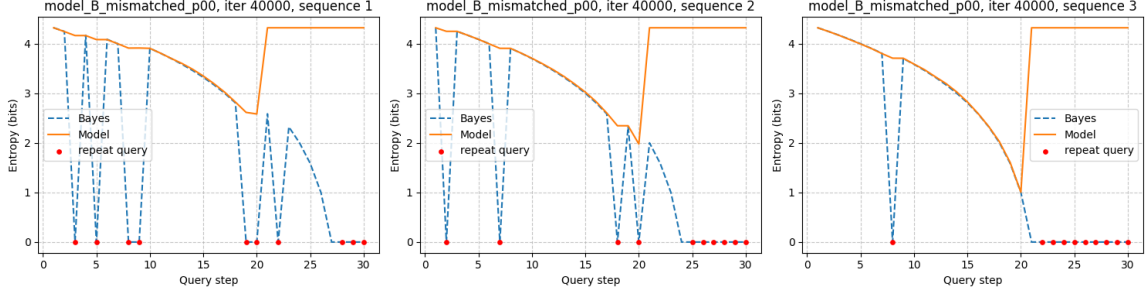


Figure 5: Per-sequence entropy trajectories for Model B prior to its transition (iteration 40,000). The model ignores early repeats and collapses to maximum uncertainty (uniform distribution) when repeats are forced late in the sequence.

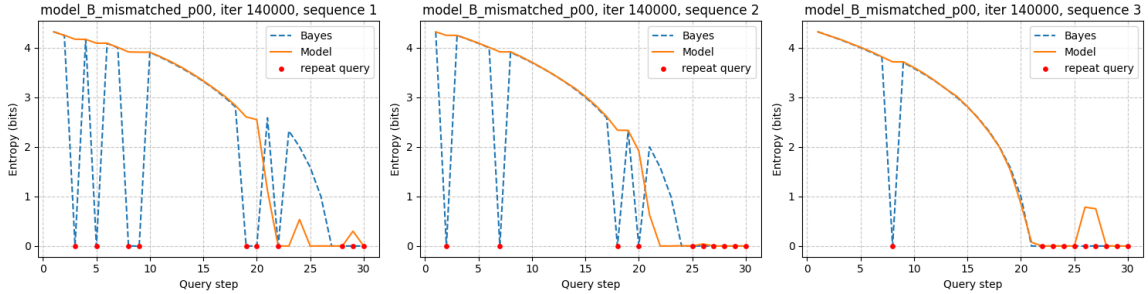


Figure 6: Per-sequence entropy trajectories for Model B at iteration 140,000. The model still fails to track early repeats but attempts to copy from step 21 onward via a positional shortcut.

The pathological nature of this shortcut is further supported by Model D (Figure 7). Its entropy trajectory completely fails to collapse on repeated queries across the entire sequence. Instead, it treats every repeat as a novel query, continuing to trace the novel-query entropy curve $\log_2(V - n_{\text{used}})$. This visually demonstrates that without early repeats, the model completely lacks associative retrieval capabilities. However, Model D is a perfect hypothesis eliminator on the distribution it was trained on.

The trajectory of Model D suggests that the sawtooth pattern reported in Agarwal et al. [1] cannot be a natural mathematical consequence of out-of-distribution generalization, but rather could be the result of an unstated experimental discrepancy.

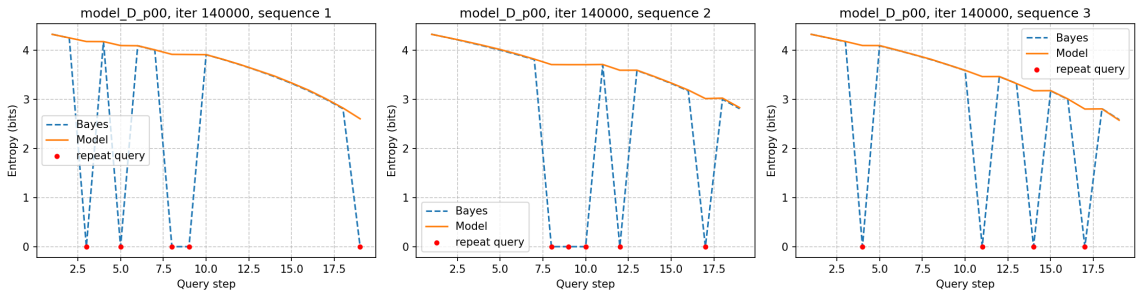


Figure 7: Per-sequence entropy trajectories for Model D at iteration 140,000. Completely isolated from repetitions during training, it treats every repeated query as a novel one.

4.3 Geometric Diagnostics

We track the key orthogonality and QK entropy (in nats) introduced in Section 2.4 across the same checkpoint schedule used for the calibration curves for all four models.

Table 6: Geometric diagnostics for Model A ($p_{\text{repeat},\text{train}} = 0.3$) at key training checkpoints.

Iteration	Key orthogonality (L0)	QK entropy (L1)	QK entropy (L3)	QK entropy (L5)
500	0.302	1.912	1.597	1.833
3,000	0.369	1.909	2.239	2.265
20,000	0.199	1.788	1.335	1.418
60,000	0.140	1.799	2.132	1.267
140,000	0.232	1.867	1.979	1.331

Table 7: Geometric diagnostics for Model B ($p_{\text{repeat},\text{train}} = 0.0$) at key training checkpoints.

Iteration	Key orthogonality (L0)	QK entropy (L1)	QK entropy (L3)	QK entropy (L5)
500	0.554	1.823	1.828	1.932
3,000	0.512	1.352	1.532	1.549
20,000	0.477	2.078	1.190	0.801
60,000	0.469	1.511	1.078	0.722
140,000	0.429	1.110	0.817	0.909

Table 8: Geometric diagnostics for Model C ($K = V - 1$, $p_{\text{repeat},\text{train}} = 0.3$) at key training checkpoints.

Iteration	Key orthogonality (L0)	QK entropy (L1)	QK entropy (L3)	QK entropy (L5)
500	0.270	1.485	1.553	1.780
3,000	0.368	1.612	1.490	1.610
20,000	0.200	1.303	1.431	1.503
60,000	0.142	1.395	1.454	1.306
140,000	0.123	1.404	1.453	1.351

Table 9: Geometric diagnostics for Model D ($K = V - 1$, $p_{\text{repeat},\text{train}} = 0.0$) at key training checkpoints.

Iteration	Key orthogonality (L0)	QK entropy (L1)	QK entropy (L3)	QK entropy (L5)
500	0.871	1.550	1.668	1.723
3,000	0.809	1.296	1.321	1.339
20,000	0.830	1.834	0.519	0.646
60,000	0.846	1.925	0.487	0.600
140,000	0.777	1.829	0.987	1.086

Key orthogonality. The models exhibit distinct geometric dynamics in Layer 0 depending on their training regime. For Models A and C, key orthogonality begins tightening concurrently with their behavioral phase transition (visible from evaluation metrics), dropping to around 0.20 by iteration 20,000. However, it continues to decrease gradually beyond this point, reaching a minimum near 0.12–0.14 much later in training. This indicates that the geometric reorganization of the hypothesis frame is not instantaneous or tightly coupled to the initial behavioral transition, but rather undergoes a slower and continuous refinement. Moreover, Model A key orthogonality

increases in late checkpoints, indicating a geometric relaxation in the hypothesis space that does not compromise the correct inference of the model.

Conversely, models B and D fail to construct this orthogonal framework, but do so in different ways. Model B’s key similarities remain noisy, fluctuating between 0.43 and 0.55. Model D has a key cosine similarity that remains completely flat and exceptionally high (≈ 0.80) throughout all 140,000 iterations. Because Model D is strictly isolated from repetitions, it has absolutely no incentive to differentiate key vectors for associative retrieval. It achieves optimal cross-entropy only through hypothesis elimination. Therefore, in our experiments the orthogonal hypothesis frame emerges only in models whose training distribution includes early repetitions and have developed an associative retrieval mechanism.

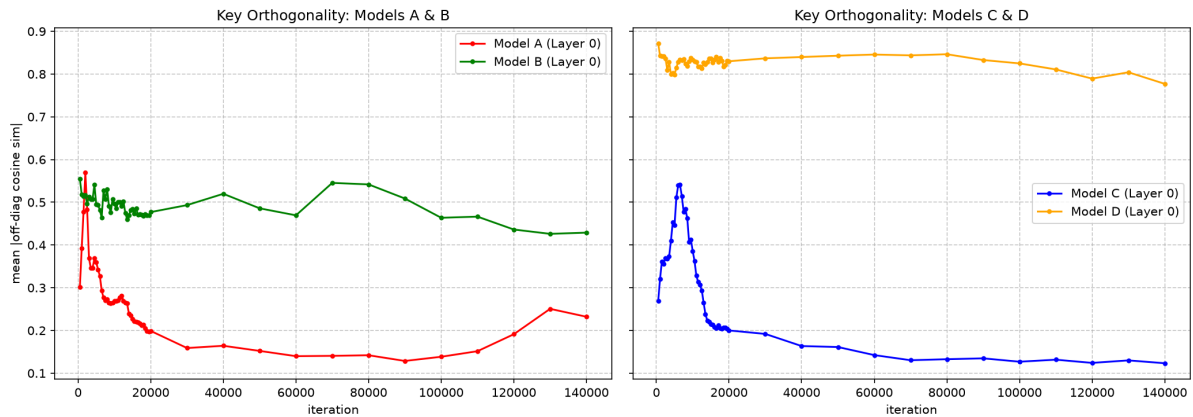


Figure 8: Mean absolute off-diagonal key cosine similarity (Layer 0) against training iteration for all models.

QK sharpening. Following the reference paper’s framing of geometric hypothesis elimination, we treat Layer 5 as the endpoint at which the cumulative effect of middle-layer QK sharpening should be most visible. Across all layers, all four models start with comparable entropy values during early initialization (≈ 1.5 – 2.0 nats). However, as training progresses, their trajectories diverge entirely depending on the repetition regime.

Models A and C develop a stable computational hierarchy: superficial layers (Layer 1) maintain an elevated entropy (≈ 1.8 nats) to preserve a broad contextual overview, while intermediate layers (Layer 3) remain high and flexible, reflecting an active phase of hypothesis exploration. Only in the deep representations (Layer 5), the models smoothly sharpen and stabilize at lower entropy values (≈ 1.3 nats). This structured profile indicates that Models A and C successfully distribute their attention across valid hypotheses in a probabilistic manner, reflecting the Bayesian hypothesis elimination framework regardless of the query horizon.

In contrast, Models B and D exhibit a strong collapse in deep-layer entropy. Model B’s Layer 5 entropy crashes deeply to ≈ 0.2 nats around iterations 40,000–50,000. Similarly, Model D drives its Layer 3 and Layer 5 entropy down to ≈ 0.4 – 0.6 nats between iterations 20,000 and 60,000. In Model B, this sharpening is consistent with the adoption of a positional shortcut, while in Model D it accompanies successful hypothesis elimination without associative retrieval. Thus, low QK entropy alone does not uniquely identify the generalized Bayesian mechanism. This geometric signature supports the interpretation that the transitions observed in these models represent the adoption of degenerate shortcuts rather than the acquisition of the generalized algorithm.



Figure 9: QK cosine-similarity row entropy (nats; lower = sharper) at Layer 1, Layer 3, and Layer 5, for all models.

Manifold Unfurling. Following the methodology of Agarwal et al., we extract the final-layer block attention outputs at all prediction positions and project them into a 3D space using Principal Component Analysis (PCA) for 200 different bijections. Figures 10 and 11 visualize these embeddings at early (500), mid-transition (20,000 for Models A and C, and 60,000 for Models B and D), and late (140,000) checkpoints, colored by the analytic Bayes posterior entropy.

For Models A and C, the unfurling of the value manifold is clear through the checkpoints. Indeed, at the early checkpoint, the representation space is collapsed, with high-entropy and low-entropy states mixed. By the mid-transition, distinct clusters begin to emerge. At the late checkpoint, the representations form a smooth low-dimensional structure that correlates with posterior entropy, with representations flowing smoothly from uncertainty to certainty.

Conversely, Models B and D fail to develop this structure. While Model B shows some clustering progression from the early to mid checkpoints, its late-stage representation space remains fragmented, with low-entropy states scattered across different regions. Model D also exhibits an unstructured and disjointed representation space at the late checkpoint.

This geometric fragmentation provides further visual evidence of algorithmic failure in Models B and D. Because they were never forced to continuously update beliefs and execute associative retrieval across all sequence positions during training, they never constructed a coherent mathematical manifold where to perform precision refinement. The fragmented representation structure is consistent with the observed failure to generalize associative retrieval to repeated queries.

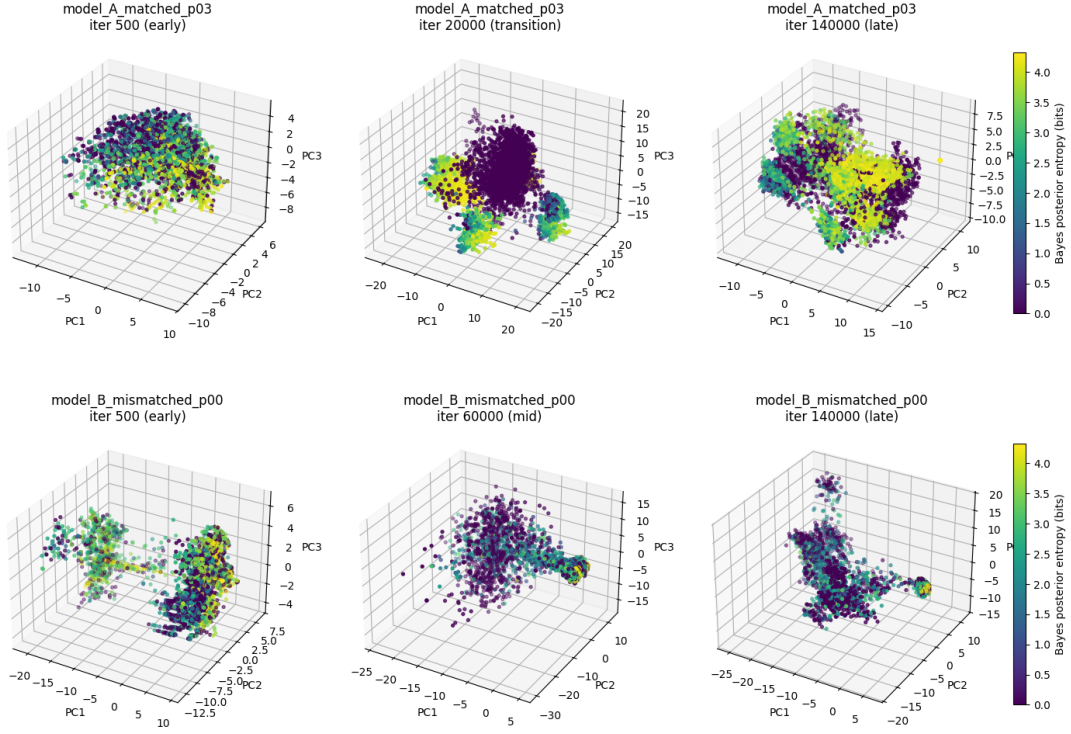


Figure 10: PCA projection of final-layer block attention outputs, colored by analytic Bayes posterior entropy (bits), at different checkpoints, for both Model A (top) and Model B (bottom).

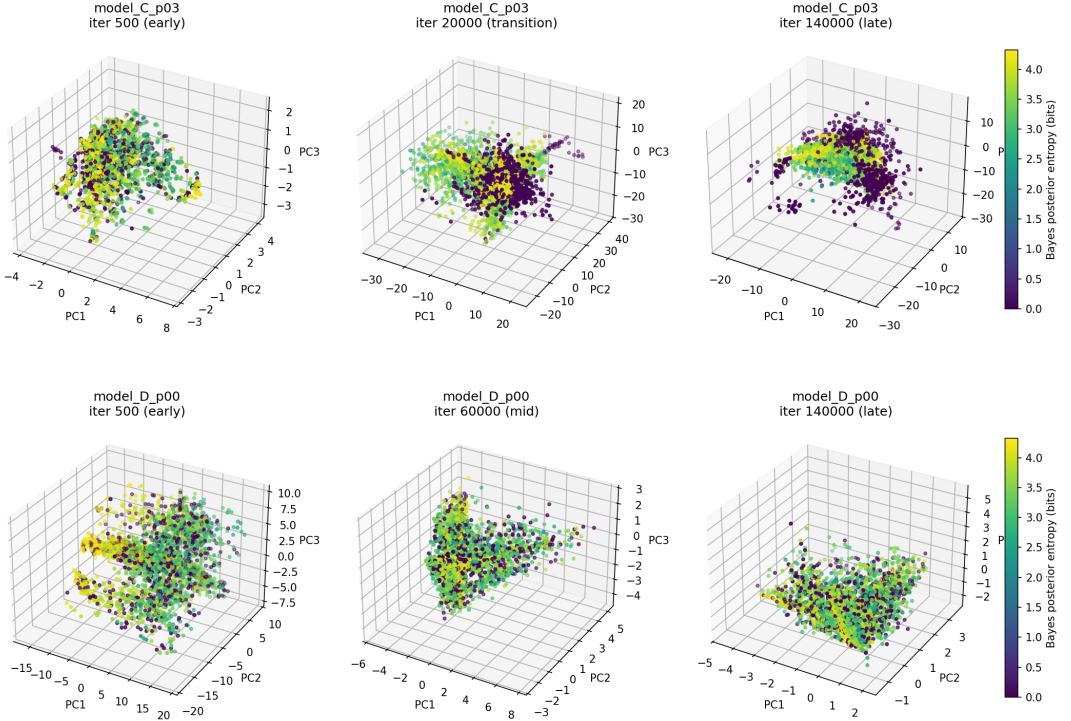


Figure 11: PCA projection of final-layer block attention outputs, colored by analytic Bayes posterior entropy (bits), for Model C (top) and Model D (bottom).

5 Discussion

Our results evaluating the reference paper’s three-stage mechanism suggest a fundamental caveat: the geometric machinery of Bayesian inference is not guaranteed to emerge outside the support of the training data, as implied by Theorem 1 and its remark.

In a repeated query evaluation framework, Bayesian geometric signatures are observed by Models A and C, which learn a generalized, position-invariant retrieval mechanism regardless of the query horizon. Foundational binding, progressive elimination, and precision refinement are absent in the other models. Model B develops a pathological geometric signature, adopting a fragile positional heuristic optimized for the forced late repeats present in its $K = 1.5V$ training data. Conversely, Model D, being isolated from repetitions during training, fails to perform associative retrieval. The population-optimum theorem fixes q^* only on the training data distribution and is therefore uninformative about configurations that never occur during training. These results highlight that good in-distribution performance does not by itself establish algorithmic generalization to structurally different inputs. Therefore, the reference paper’s reported per-sequence diagnostic cannot be a direct mathematical consequence of their framework and likely originates from an unstated experimental discrepancy.

Finally, the models exhibit an asymmetry in the timing of their transitions. In Models A and C, the behavioral and geometric phase transitions do not occur sharply together: geometric refinement continues well after the behavioral transition (by roughly 70,000 iterations). In contrast, Model B’s geometric adjustments coincide with its behavioral transition around iteration 60,000, while Model D shows no functional transition. This suggests that a sharp simultaneity of behavioral and geometric changes may reflect a local and opportunistic adjustment rather than the formation of a generalizing hypothesis framework.

Methodological Limitations. This work has some methodological boundaries. Primarily, our results rely on a single training seed for each model due to computational constraints, and future work should evaluate multiple runs. Furthermore, while our 2×2 design compares the role of the query horizon K , future work could test variations in the domain size V to observe how the timescale of these phase transitions scales with vocabulary complexity.

6 Conclusion

This study provides a critical reproduction of the Bayesian bijection wind tunnel proposed by Agarwal et al. [1]. By introducing a 2×2 experimental design crossing query horizons ($K = 1.5V$ versus $K = V - 1$) with training repetition regimes, our framework was designed to probe an inconsistency in the original paper’s claims.

If evaluation includes repeated queries, following Agarwal et al. [1], only models exposed to repeats distributed across the full sequence during training (Models A and C) successfully track the exact Bayesian posterior predictive distribution. Models never exposed to early repeats (Models B and D) fail, either relying on a degenerate positional shortcut confined to the forced-repeat region (Model B), or failing to develop associative retrieval at all (Model D). However, if evaluation is conducted without repeated queries, thus matching the training support of all models, then all models perform well, achieving near-zero MAE and KL divergence (see calibration metrics in Tables 2–5).

Ultimately, our results are consistent with the population-optimum formulation of Theorem 1, but also clarify its limitation: the theorem determines the optimal predictor on the training distribution, not the algorithm learned by the model, nor its behaviour outside that distribution. A model may therefore achieve the Bayesian predictive posterior during in-distribution evaluation, while relying on a representation or heuristic that fails when the evaluation introduces structures

absent from training. The emergence of the geometric signatures described by Agarwal et al. [1], as well as the emergence of Bayesian updating itself, must therefore be evaluated in relation to the precise support and structural constraints of both the training and evaluation distributions.

7 AI Usage

This project was carried out in accordance with the Physics of Data AI policies. Specifically, Generative AI tools (including Claude model Sonnet 5 and Gemini) were utilized to assist in developing sections of the Python code, particularly concerning the adaptation of minGPT for this project, as well as for text revision and information retrieval.

References

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